
Introduction to medical parasitology and Nemathelminthes

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✓ Contents

- ✓ Introduction to medical parasitology
- ✓ Nematelminthes
- ✓ **Platyhelminthes**
- ✓ **Introduction to protozoa**
- ✓ **The Amoeba**
- ✓ **Flagellate protozoa**
- ✓ **Apicomplexa**

Outline

- ▶ Definition, scope and importance of parasitology
- ▶ Types of parasites, hosts and vectors
- ▶ Sources of infection, Mode of transmission, Routes of transmission
- ▶ General life cycle of parasites (Direct and Indirect life cycle)
- ▶ Classification and features of nematodes
 - ▶ Transmission, life cycle, clinical features, laboratory diagnosis, treatment, prevention & control

✓ Learning objectives

At the end of this session the students will be able to:

- ▶ Discuss the definition, scope and importance of parasitology
- ▶ Explain types of parasites, hosts and vectors
- ▶ Discuss the source of infection, mode of transmission, route of entry of parasites and the general life cycles of parasites
- ▶ Discuss the transmission, life cycle, clinical features, laboratory diagnosis, treatment, prevention & control measures of each nematodes

✓ Introduction to medical parasitology

▶ **Parasitology:** **parasite + logy** (the study of parasite)

▶ **Parasites:**

▶ Greek: **para + sitos (besides / alongside + food)**

▶ Literally means person eating at another's table

▶ are living organisms that use one / more than one living animal or plant species as a host for their nourishment and survival

▶ Study of those living closely to each other or phenomenon of dependence of one living organism with the other

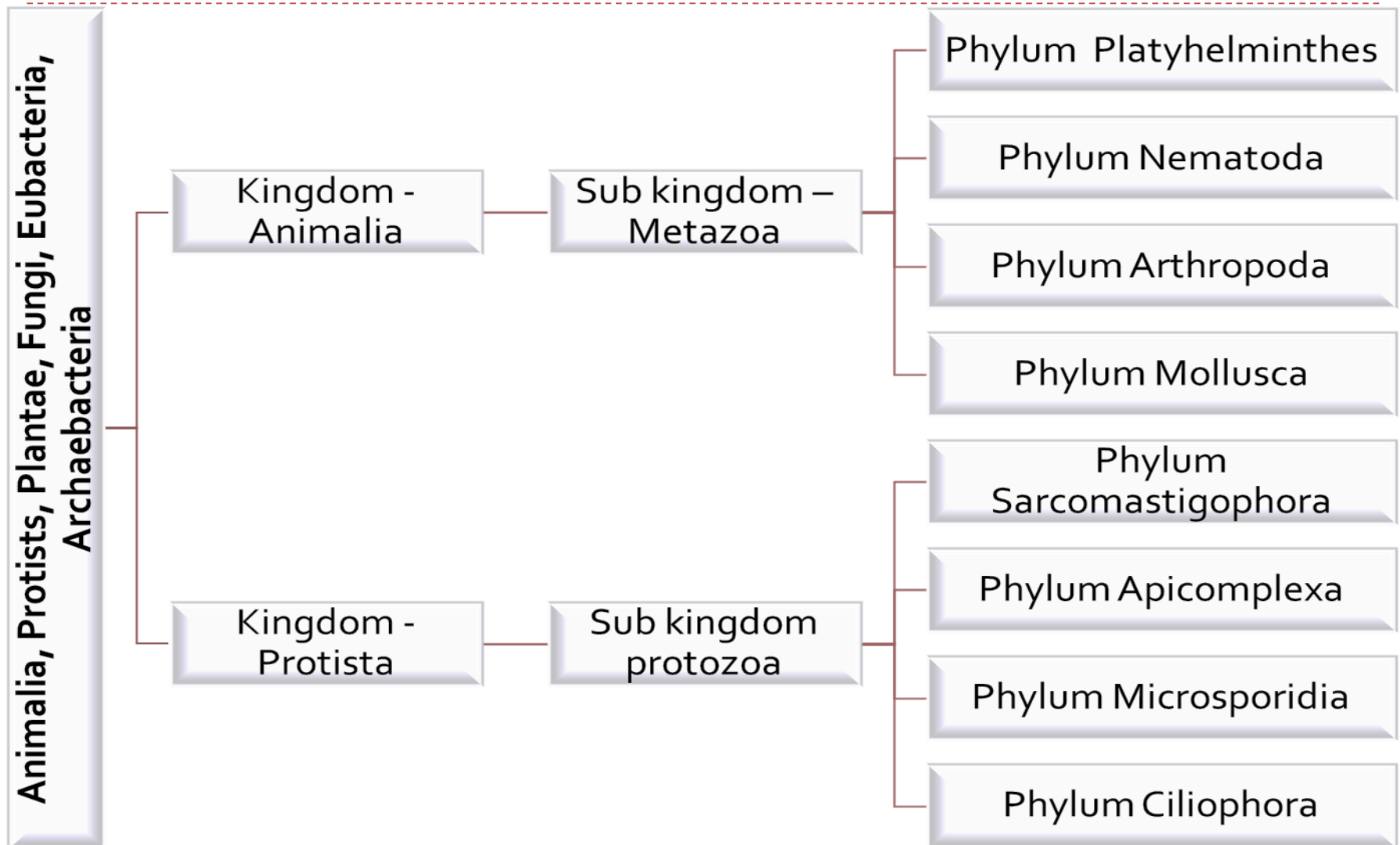
Definition Cont'd....

- ▶ **Hosts:** are organisms which harbors the parasite, provides nourishment and shelter, is relatively larger than the parasite and is the injured partner
- ▶ **Vector:** is an organism (usually an arthropod) which transfers infective forms of a parasite from one host to the other.
- ▶ **Therefore, Medical parasitology is:**
 - ▶ The study of the parasites of **man** and their **medical consequences**

✓ Scope of Medical Parasitology

- ▶ Medical parasitology is a broad discipline incorporating:
 - ▶ **Medical protozoology**: study of protozoa (single celled)
 - ▶ **Medical helminthology**: study of metazoan helminths (multi-celled)
 - ▶ **Medical entomology & malacology**: study of arthropods & mollusks
- ▶ The study of parasitism is **also interdisciplinary** with
 - ▶ Systematics and phylogeny, ecology, embryology, physiology,
 - ▶ Biochemistry, immunology, pharmacology, nutrition, biology, etc.

✓ Classification of parasites



✓ Importance of Medical Parasitology

- ▶ Are still challenging problems in developing countries, we need to know parasites for their prevention and control
- ▶ Distributions of parasitic diseases are evolving and changing due to globalization
- ▶ Majority of Neglected tropical diseases (NTDs) to which WHO pays great attention are parasitic infections
 - ▶ —~~Malaria~~, Schistosomiasis, Filariasis, Leishmaniasis, STHs, Schistosomiasis,
 - ▶ Trypanosomiasis, Lymphatic filariasis, Taeniasis/cysticercosis,
 - ▶ Echinococcosis, Dracunculiasis, Onchocerciasis, food borne trematodes

Parasitic infections are still challenging among other infectious diseases....

- ▶ Affects underprivileged or poor communities, thus neglected
- ▶ Chronic nature of diseases, higher morbidity than mortality
- ▶ Lack of effective vaccine, no or few effective drug or Drug resistance emergence
- ▶ Majorities are vector borne which **depends on Parasite, Host & Vector nature**
- ▶ Mostly affect young reproductive and both extreme age groups

✓ **Parasite – host interactions / associations**

- ✓ **Host-parasite interaction is a complex association** in which two or more different organisms live together and interact

- ▶ **Types of associations:**

- ▶ Symbiosis, Mutualism, Commensalism, phoresies, Parasitism

- ✓ **Symbiosis**

- ▶ Any two organisms living in close association one living in or on the body of the other usually for nutritional benefit
 - ▶ Both organisms benefit from the association

Association Cont'd....

✓ **Mutualism**

- ▶ Permanent and obligatory association b/n two metabolically dependent organisms benefitting each other
- ▶ Termites digest wood (cellulose) b/c their intestinal protozoan flagellates synthesize cellulase

✓ **Phoresis** “means to carry in Greek”

- ▶ The phoront, usually the smaller organism, is mechanically carried by the other usually larger organism, the host.
- ▶ No physiologic interaction or nutritional dependency is involved.

Association Cont'd....

✓ **Commensalism:**

- ▶ One partner is benefited while other neither benefited nor injured
- ▶ Not involve physiological interaction & can live independently
- ▶ Spatial proximity allows the commensal to feed

✓ **Parasitism**

- ▶ the parasite, either harms its host or in some sense lives at the expense of the host, is the benefited partner
- ▶ a parasite, the smaller of the two, is physiologically dependent

✓ Types of Parasites

According to their habitat

I. Endoparasite

- ▶ Inside, under the surface or deep in the body
 - ▶ Tapeworms, flukes, protozoans

II. Ectoparasite

- ▶ Stays on outside surface of the host
 - ▶ Lice, ticks, fleas

Based on dependency on the host

I. Obligate Parasite

- ▶ Requires finding and invading the host to complete its life cycle
- ▶ Most of the parasites are obligate parasites

II. Facultative Parasite

- ▶ Does not require a host.
- ▶ May become parasitic and even fatal if given the chance
- ▶ Example: free living amebas: *Naegleria fowleri*

Based on amount of time spent on host

I. Permanent Parasite

- ▶ Lives entire adult life stage on or in a host
- ▶ Usually Endo-parasites, Example: helminthes, protozoans

II. Temporary or intermittent Parasite

- ▶ Spends only a short time on a host
- ▶ Usually Ecto-parasites, Example: mosquito, bed bug

According to their Pathogenicity

I. Pathogenic parasite

- ▶ Cause morbidity and mortality of host

II. Non-Pathogenic

- ▶ Can't harm the host

III. Opportunistic parasites

- ▶ Cause disease depending immune status of the host

Based on their life cycle

I. Monoxenous parasites:

- ▶ Those with simple or direct life cycles & needs one host only

II. Heteroxenous parasites:

- ▶ Those with complex & indirect life cycles, needs 2 or more IH

Heterogenetic Parasites:

- ▶ Life cycles of some parasites have regular alternation of sexual & asexual reproductions. E.g. *Plasmodium spp*, *Strongyloides*

Based on host ranges

I. Euryxenous parasites

- ▶ Those with a broad host range or infect a large group of host spp

II. Stenoxenous parasites

- ▶ Those with a narrow host range, infect only one species, or closely related *spps*

- ✓ **Parasites – Host specificity** is determined by immunological, genetic, physiological, and/or ecological factors

Based on location of developmental stages in the host

I. Aberrant parasite

- ▶ Found in **locations** in the host where do not occur normally
- ▶ No further development but simply wander until their death
- ▶ *Ascaris larvae* may migrate to the brain

II. Incidental or accidental parasite

- ▶ Occurs in **hosts** where it does not occur normally
- ▶ Fasciola incidental if found in man's liver.

✓ **Types of hosts**

✓ **Definitive host**

- ▶ Host where the adult stage of parasites live & sexual reproduction takes place
- ▶ Normally the larger of the hosts, usually a vertebrate

✓ **Intermediate host**

- ▶ Host where sexually immature or larval stage of a parasite live & asexual multiplication takes place

Types of hosts cont'd...

✓ **Paratenic or Transport Host**

- ▶ No development or interaction occurs
- ▶ Parasite remain alive & infective to another host

✓ **Accidental or Incidental Host**

- ▶ Parasite is in the “wrong” or unusual host species
- ▶ Parasite usually wanders around and causes great damage because it doesn't know where to go then dies mostly
- ▶ No adaptation between host and parasite

Types of hosts cont'd...

✓ **Reservoir Host**

- ▶ Any organism in which an infectious agent normally lives and multiplies until transmitted to the main or primary host
- ▶ Serves as a source from which more than one host can be infected without injury to itself

✓ **Carrier host**

- ▶ Host that harbors parasites without showing any clinical symptom
- ▶ Can be an important source of infection in epidemiology

✓ Types of vectors

A. Biological vectors:

- ✓ Parasite dev't and multiplication before transfer to other host
 - ✓ **Propagative:** Multiplication but no change in form of parasite
 - ✓ **Cyclo propagative:** Both multiplication and cyclical dev't
 - ✓ **Cyclo developmental:** Cyclical dev't but no multiplication

B. Mechanical vectors:

- ✓ A transport host with no parasitic development or reproduction



✓ **Transmission of parasites**

□ **Key link factors** for parasite transmission are:

- ✓ **Source** of infection or exposure,
- ✓ **Mode** and **route** of transmission and
- ✓ Presence of **Susceptible host**

1. **Source of infection or exposure**

- ▶ **Primary** sources of Exposure to Parasitic Infections are **infected persons, carriers and animals**

Source of infection cont'd...

Human beings:-

- ▶ A person his/her clothing, bedding or the immediate environment that he/she contaminated

Autoinfection: -

- ▶ *S. stercoralis*, *E. vermicularis*, *H. nana* & *T. solium*, *C. parvum*

Animals (domestic or wild animals harboring the parasite

- ▶ Dogs- the hydatid cyst caused by *E. granulosus*

Source of infection cont'd...

Contaminated soil

- ▶ Soils polluted with human excreta is source for STHs

Blood sucking arthropods:

- ▶ Malaria: *A. mosquito*, *Leishmania*: sand flies

Contaminated water with infective stages:

- ✓ viable cysts, Cercarial stages, Cyclops containing larva,

Raw or Insufficiently cooked meat of pork, beef and fish

- ▶ *Trichinella spiralis*, *Taenia* species, *D. latum*

2. Mode and route of Transmission

Mode of transmission

✓ means how parasite invades human body

1. Direct mode of Transmission

- **Horizontal:** direct contact with infective stage thru Feco-oral route, Sexual intercourse, skin contact, transfusion
- **Vertical:** from mother to child congenitally, trans placental, Trans mammary

2. Indirect Mode of Transmission

- ▶ In parasites with complex life cycle, requires one or more IH

Route of Transmission

- ✓ Specific **entrance way through which** the parasite invades the human body
 - ▶ **Ingestion of infective stage of parasites:** In food, water contaminated with faeces, In raw or undercooked meat,
 - ▶ **Through Insect Bite:** filarial worms, Plasmodium sp.

Mode and route cont'd...

- ▶ **Penetration of skin when in contact with:** Fecally polluted soil, Water containing infective stages
- ▶ **Sexual Contact:** Ex. *Trichomonas vaginalis*
- ▶ **Trans mammary:** Ex. *S. stercoralis*, *T. gondi*
- ▶ **Inhalation of contaminated air:** Ex. *E. vermicularis*, *P. carinii*
- ▶ **Trans placental:** Ex. *T. gondii*
- ▶ **Kissing:** Ex. *Entamoeba gingivalis*, *Trichomonas tenax*

✓ Life Cycles of parasites

- ▶ Describes the cycle of development of the parasite,
- ▶ This may involve passing through a number of developmental stages & environment, Parasitic and non-parasitic stages.
- ▶ Ova (non parasitic) → Larva (parasitic) → Adult (parasitic stage)
- ▶ **Major phases** in the life cycle of parasites:
 - ▶ **Growth and development (maturation), Reproductive** (sexual and asexual) and **Transmission phases.**
 - ▶ All are vitally important for the successful survival of parasite.

Life cycle cont'd...

- ✓ Depending on number of hosts required to complete its cycle

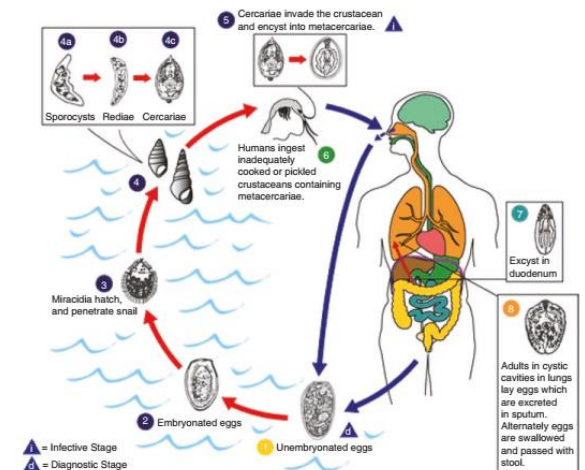
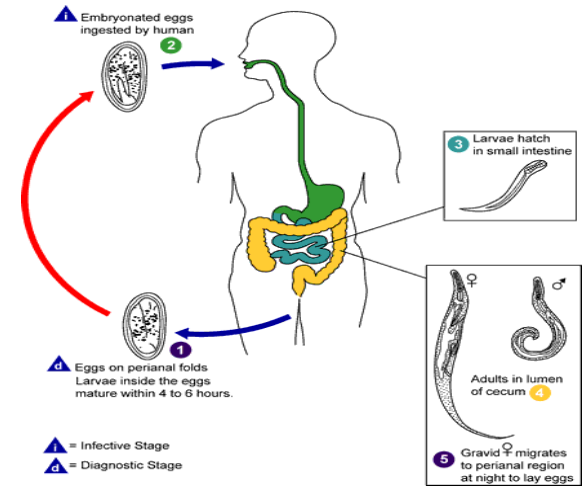
A. Simple or Direct Life Cycle (monoxenous)

- ▶ Only one host is required to complete its cycle
- ▶ The parasite often spends most of its life, usually as an adult, and where it reproduces
- ▶ Transmitted from one host to another through the air, by a fomite, or in contaminated food or water from soil.

Life cycle cont'd...

B. Indirect or complex life cycles (heteroxenous)

- requires 2 or more hosts (a vector or intermediate host) to reproduce or grow in
- Frequently this may involve passing through a number of developmental stages & Evt.



► **Nemathelminthes**

✓ General features of Nematodes

- ▶ Round in their cross-section and non segmented
- ▶ Possess a shiny cuticle (smooth/spined/ridged)
- ▶ Have a separate sexes (**Dioecious**), male smaller with curved tail
- ▶ Can be **oviparous/ ovoviviparous/ viviparous**
 - ▶ Egg (ova) → Larva (Four stages of molting, L1 to L4) → Adult
- ▶ Digestive system is complete, possess mouth, oesophagus & anus
- Mouth is surrounded by lips or papillae



✓ Classification of Nematodes

1. Intestinal nematodes

Small intestine

- ▶ *Ascaris lumbricoides*
- ▶ Hook worm
- ▶ *Strongyloides stercoralis*

Large intestine

- ▶ *Trichuris trichuria*
- ▶ *Enterobius vermicularis*

2. Blood & tissue nematodes

▶ Filarial worms

- ▶ *Wuchereria bancrofti*
- ▶ *Brugia malayi*
- ▶ *Onchocerca volvulus*
- ▶ *Loa loa*
- ▶ *Trichinella spiralis*
- ▶ *Draconculis medinensis*

✓ Intestinal nematodes

✓ General features

- ▶ Live in gastro-intestinal tract
- ▶ In humans, often spread by poor hygiene related to feces
- ▶ Most species are geo-helminths (soil transmitted)
- ▶ Female worms are **oviparous**
- ▶ Humans are the only or major host of intestinal nematode (direct life cycle)

Intestinal nematodes cont'd....

▶ **Transmission:**

- ▶ Ingestion of infective egg
- ▶ Larva penetrating skin

▶ **Laboratory diagnosis:**

- ▶ Egg in feces (most often)
- ▶ Larva in feces
- ▶ Recovering egg in the skin around the anus
- ▶ Occasional adult worms: *A. lumbricoides*, *E. vermicularies*

✓ *Ascaris lumbricoides*

- ✓ **Common name:** Large Intestinal round worm
- ✓ **Disease known as** *Ascariasis*
- ✓ Occurs globally with prevalence of 25% (0.8 - 1.22 billion people) infected worldwide and 60,000 death annually
- ✓ In Ethiopia (prevalence ranges from 6 – 77%)
- ✓ The **most prevalent** and occur in highest rate in children
- ✓ Distributions affected by altitude and climate

A. lumbricoides cont'd....

➤ **Habitat**

▶ **Adult:** In the small intestine

▶ **Egg:** In the faeces

✓ Extremely resistant to adverse environmental condition and chemicals

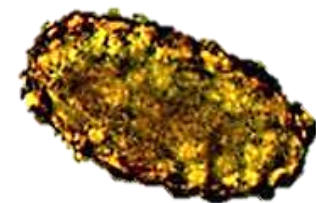
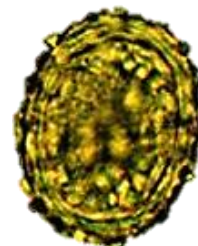
✓ Remains viable in soil and dust for up to 10 years

➤ **Morphology**

▶ **Adult:** Color pinkish, cylindrical with tapering ends

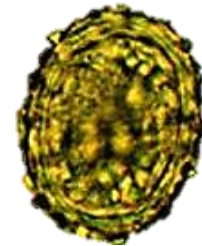
A. lumbricoides cont'd....

- **Male:** Size: about 15-30cm curved tail, two copulatory spicules
- **Female:** Size 20-35cm, with a straight tail.
- The adult worm has a lifespan of 1 – 2 years
- Capable of producing 200, 000 eggs per day
- **Eggs:** fertilized and unfertilized eggs in stool



A. Fertilized Egg

- ▶ broad oval, or some times round shape, about 45 x 75µm
- ▶ thick, rough, distinct two layers, lumpy external shell,
- ▶ consists of ascaroside, chitinous layer, fertilizing membrane and smooth mammillated albuminous coat
- ▶ yellow brown bile stained



B. Unfertilized Egg

- ▶ more elongated elliptical with 80-90µm size
- ▶ brown, puffy external and thin internal shell, bile stained
- ▶ The chitinous layer & albuminous coat are thinner, without ascaroside and fertilizing membrane
- ▶ Produced by unmated female or before fertilization occurs
- ▶ Do not develop to infective stage

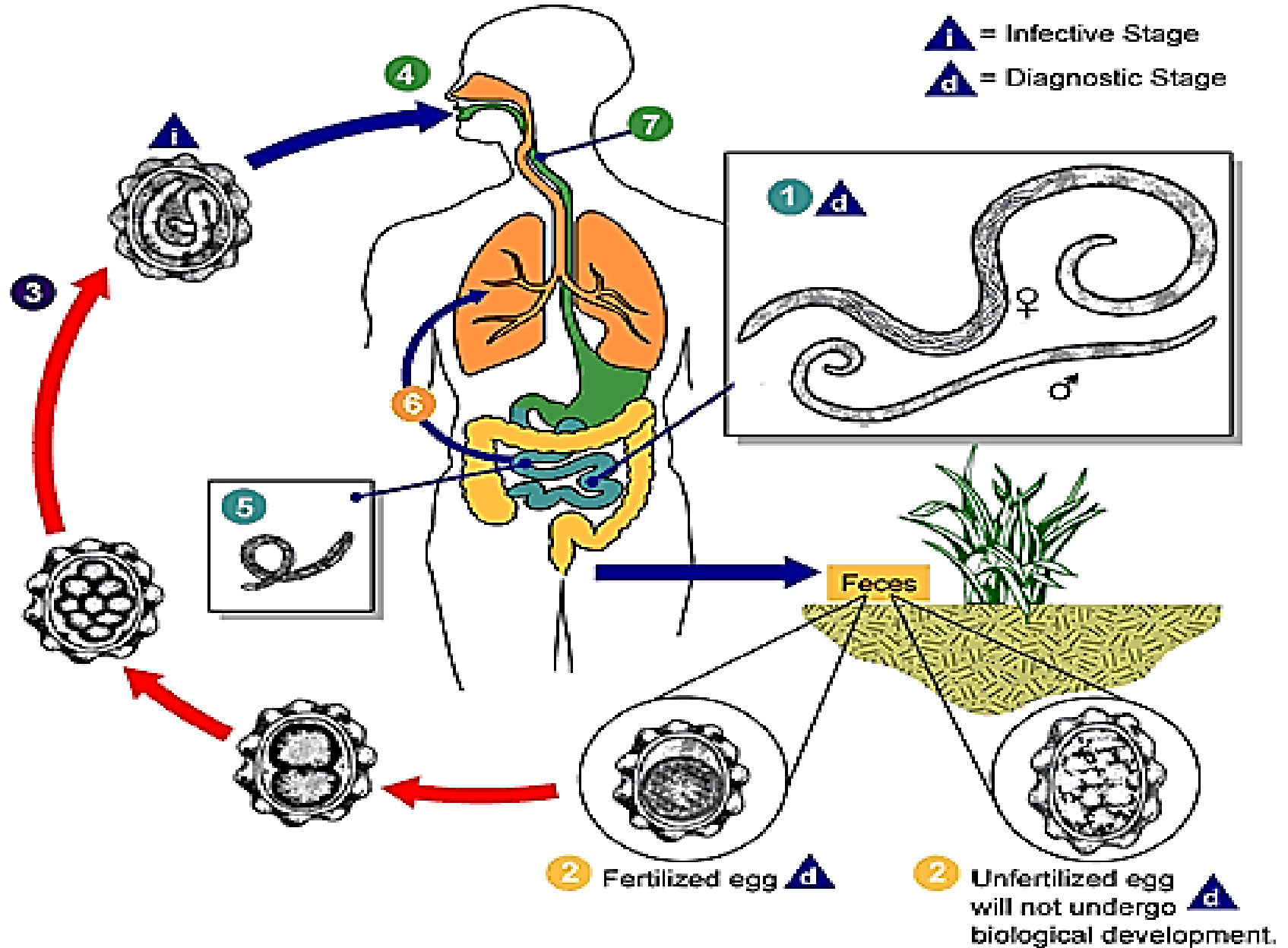


A. lumbricoides cont'd....

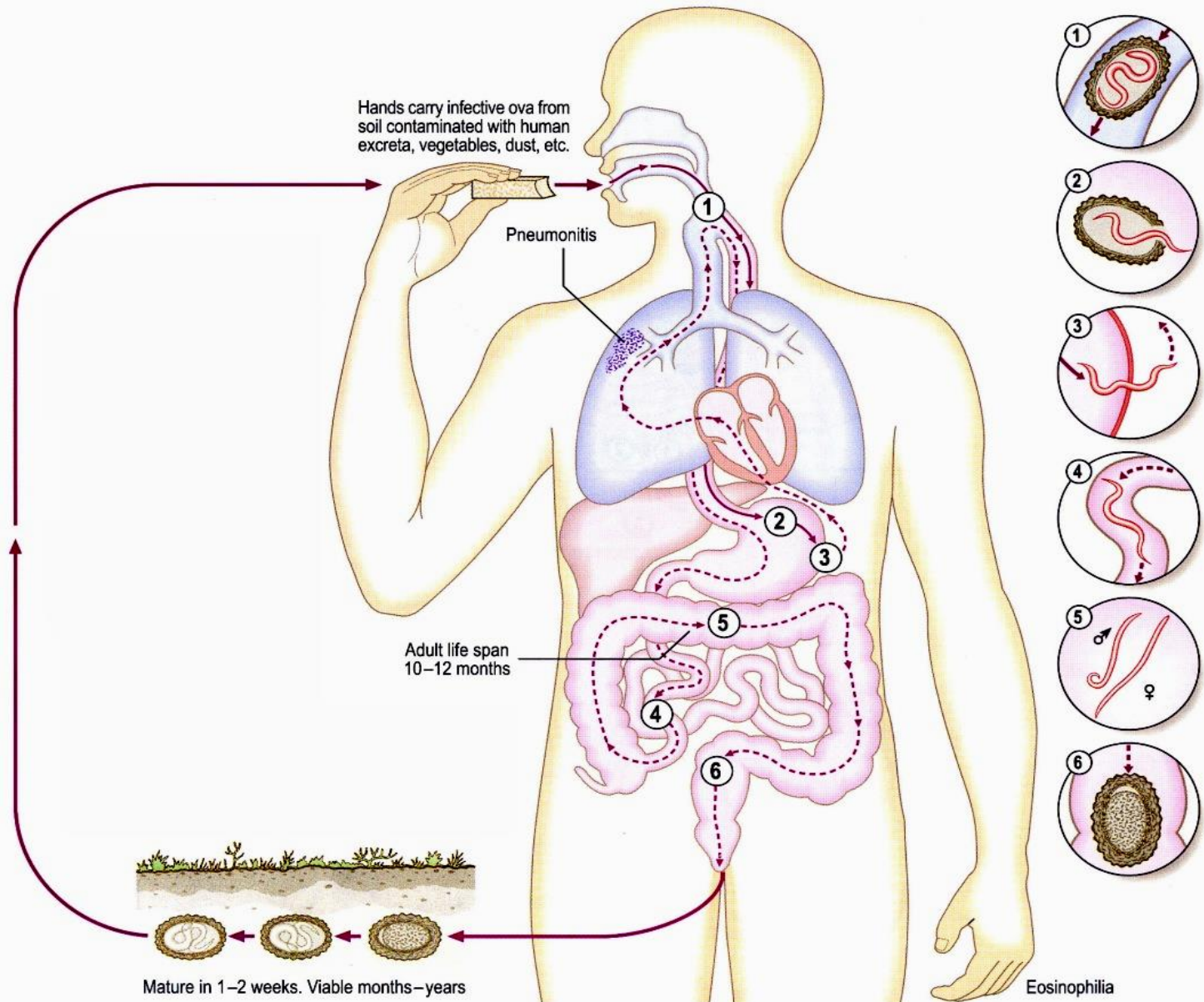
- ▶ **Transmission: A person acquire infection by:**
- ✓ Ingestion of food or water contaminated with infective stage ova containing 2nd stage larva
- ✓ Eating fecally polluted soil (Geophagia) frequently seen in children, few pregnant women
- ✓ Putting contaminated finger or toys with infective egg in to mouth
- ✓ Rarely by inhalation of eggs carried in air

✓ Life cycle

- ▶ Fully embryonated eggs are swallowed & L2 hatches in the stomach & penetrate stomach or duodenal mucosa
- ▶ L2 enter blood stream, heart & leave through alveoli into lung, then molt twice in the lungs to L3/L4
- ▶ Then move up and get swallowed reach small intestine
- ▶ Adult worms lay about 200,000 eggs/day after 2-3 months
- ▶ Eggs are shed with the feces and embryonate within 2-3 wks



A. lumbricoides cont'd....



✓ Pathogenesis

- ✓ migratory larvae causes “Verminous pneumonia”, due to lung **tissue damage**.
- ✓ Intestinal **obstruction** - too many adult worms.
- ✓ Parasite **secretes trypsin inhibitor**, prevents host from **digesting proteins**.
- ✓ **Aberrant migration** of adult worms during fever and when seeking mates
 - ✓ Common duct, Liver, Pharynx, Peritoneum cause obstruction & inflammation

✓ Clinical features

- ▶ Depends upon worm burden and sensitization
- ▶ **Lung phase of Juvenile worm: 5-6 days after exposure**
 - ▶ Pneumonitis (Loeffler's Pneumonia), hemorrhage due to penetration of lung capillaries
 - ▶ An irritating nonproductive cough & burning substernal discomfort aggravated by coughing or deep inspiration.
 - ▶ Dyspnea and blood-tinged sputum are less common.
 - ▶ Eosinophilia subsides slowly after symptomatic phase

Clinical features cont'd....

▶ **Intestinal phase: 2-3 months after infection**

- Abdominal pain, diarrhea, malabsorption of nutrient, Stunted growth, cognitive impairment, intestinal obstruction
- Wandering worm and heavy worm burden
 - Migration of an adult worm up the esophagus can provoke coughing and oral expulsion of the worm.
 - Intestinal perforation, peritonitis, blockage of bile ducts, obstruction of respiratory tract, liver abscess



✓ **Laboratory Diagnosis**

- ✓ Finding & identifying eggs in the stool (Intestinal Ascariasis)
 - ▶ Direct wet mount is adequate in moderate to heavy infection
 - ▶ Concentration technique may be used In light infection
- ▶ Adult worms occasionally passed in the stool or through the mouth or nose
- ✓ Larvae can be identified in sputum or gastric aspirate during the pulmonary migration phase

✓ Treatment

- ▶ **Albendazole:** A single dose of 400 mg or
- ▶ **Mebendazole:** 100 mg BID for 3 days or single dose of 500mg
- ▶ **Ivermectin:** 150-200mg/kg once is effective
- ▶ **Pyrantel pamoate:** 11mg/kg once, max 1g is safe in pregnancy
- ▶ **Partial intestinal obstruction:** nasogastric suction, IV fluid & instillation of piperazine through nasogastric tube
- ▶ **Complete obstruction** and its severe complications require immediate surgical intervention

✓ **Prevention and Control**

- ▶ Prevention of infection by
 - ✓ Washing hands before eating & after defecation, trimming finger, avoid eating uncooked foods such as vegetables
- ▶ Preventing soil become polluted with feces by:
 - ✓ Sanitary disposal of feces in latrines, avoid use of night soil as fertilizer
- ▶ Health information provision and treating infected individuals
- ▶ MDA programmes repeated at 3-6 month intervals (high burden)

✓ Hook Worm species

- Two major species
 - *Ancylostoma duodenale* (Old world Hook worm)
 - *Necator americanus* (New world Hook worm)
- ▶ Definitive host is human but no Intermediate host
- ▶ Are **hematophagous** or are the voracious blood feeders of all the nematodes
- ▶ *Ancylostoma duodenale* is longer, thicker and more pathogenic than *Necator americanus*

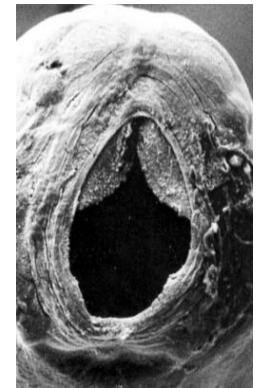
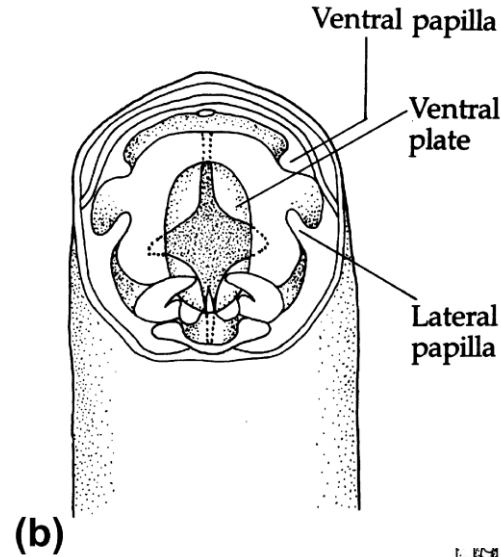
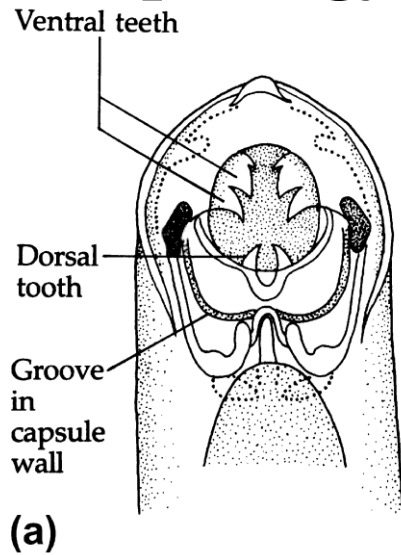
✓ Epidemiology

- ▶ Widely distributed throughout the **tropics** and **subtropics**
- ▶ More than **1 billion** people are infected world-wide
- ▶ Children, agricultural workers and miners are mostly infected
- ▶ In Ethiopia: *N. americanus* are more common **than** *A. duodenale*
- Altitude and moisture majorly affect their distribution
- Hook worm infection is **absent in low**, hot dry areas of Ethiopia and above **2500m** altitude

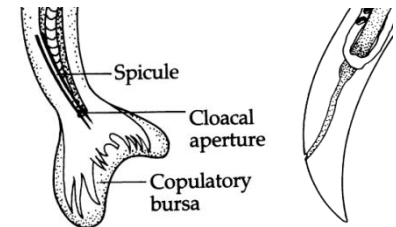
✓ Habitat

- ▶ **Adult:** small intestine: Jejunum and less often in the duodenum of man
- ▶ **Eggs:** In the feces; not infective to man
- ▶ **Two forms of larva: (Rhabditiform & filariform larvae)** free in soil and water
- ✓ **Organ of attachment to host tissue:**
 - ✓ Teeth in their buccal cavity enable their attachment to intestinal mucosa from where they suck their host's blood

✓ Morphology of Adult



16-12 Hookworm buccal capsules. (a) *Ancylostoma*. (b) *Necator*.



- Head is slightly bend (hook) and
- Mouth characteristic teeth (4 for *Ancylostoma*, 2 on each side) or plates (2 for *Necator*, 1 on each side)
- Posterior end of male is elaborated into a copulatory bursa

✓ Morphology of egg

- ✓ *A. duodenale* produce 20,000 egg/day which is twice than *N. americanus*
- ✓ **Size** : 65-40µm and **Shape**: oval
- ✓ **Shell**: very thin and appears as black line
- ✓ **Colour**: the cells inside are pale gray
- ✓ **Content**: contains an ovum which appears segmented of about 4-8 blastomeres



✓ **Transmission**

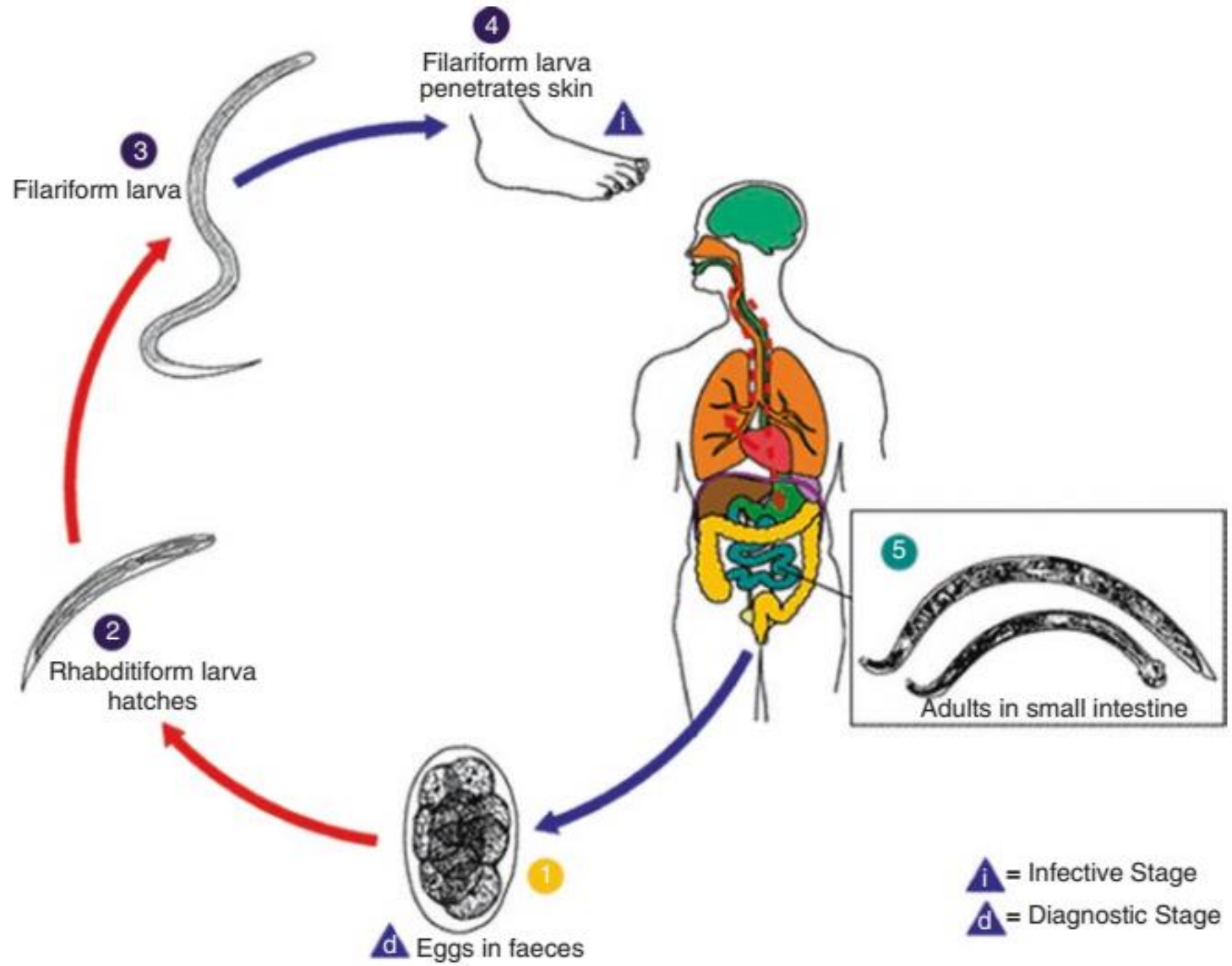
- ✓ Penetration of the skin by filariform larvae
- ✓ Ingestion of the filariform larvae present in the soil or
- ✓ Trans mammary (*A. duodenale*),
- ✓ Transplacental: rare

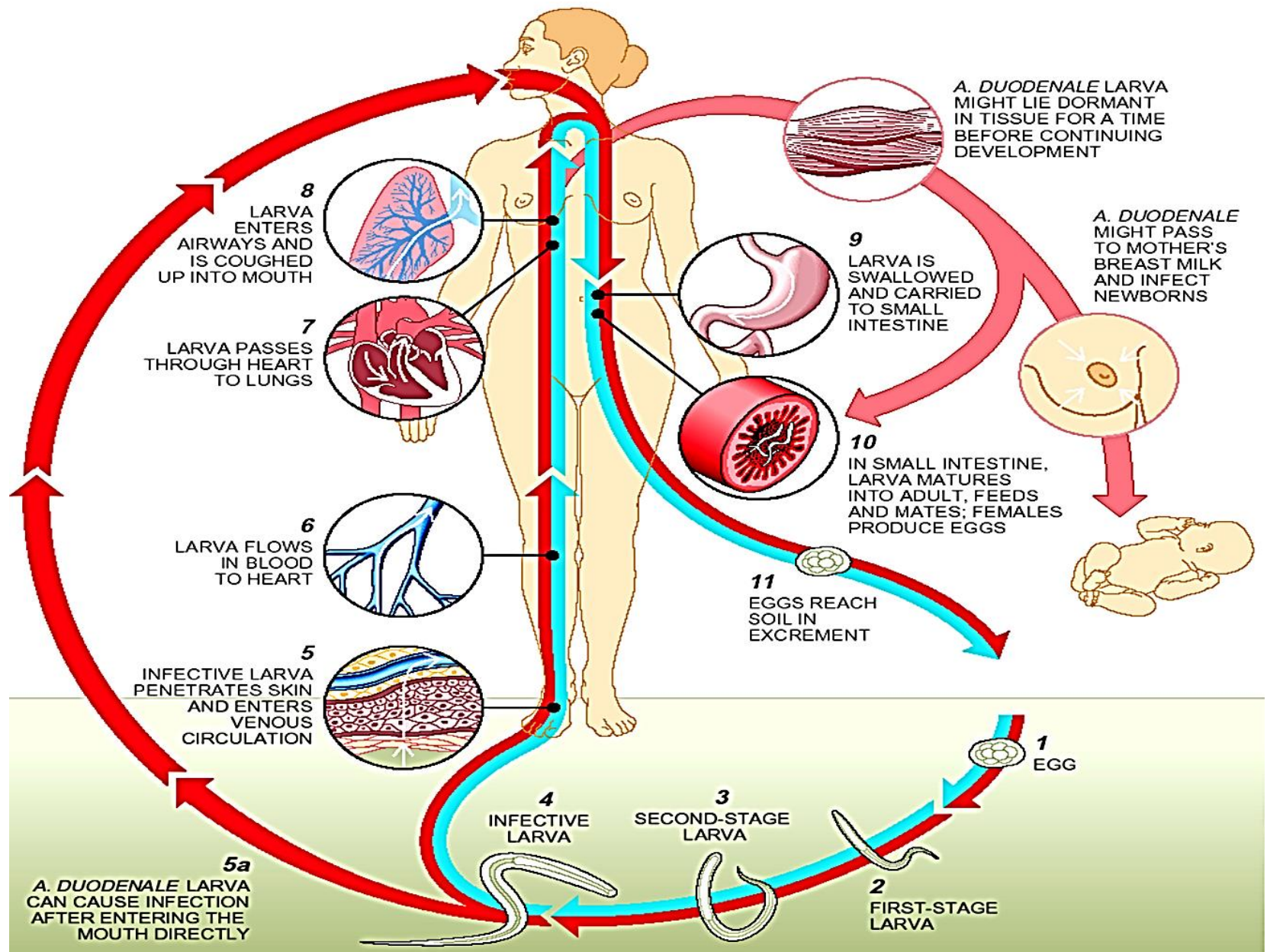
✓ Life cycle

- ▶ Eggs passed in the feces & rhabditiform larva **hatch** in 1-2 days under favorable conditions (shade, moisture, warmth)
- ▶ After 5 to 10 days (molts twice) and they become filariform **L3** larvae that are infective
- ▶ Larva can **survive** 3 to 4 weeks in favorable environmental conditions.
- ▶ On contact of host, the larvae penetrate the skin and carried through the veins to the **heart** and then to the **lungs**

Life cycle cont'd....

- ▶ They penetrate into the pulmonary alveoli, ascend the bronchial tree to the pharynx and **swallowed**
- ▶ The larvae reach the **small** intestine, where they reside and mature into adults
- ▶ They **attach** to the intestinal wall with resultant **blood loss** by the host
- ▶ The worm's mean life span is: 15 years for *A. duodenale* and 3-7 years for *N. americanus*





✓ Clinical features

- ▶ Most infections are asymptomatic (90%)
- ▶ arise from a combination of intestinal inflammation and progressive iron/protein-deficiency anemia
- ✓ **Larval migration through skin and tissue**
 - ✓ Dermatitis known as "**ground itch**",
 - ✓ Allergic reaction, papule with itching and burning sensation
 - ✓ Scratching leads to secondary infection

Clinical feature cont'd...

- ✓ **Pulmonary phase is usually asymptomatic**

- ✓ Löffler's syndrome, Pneumonitis (allergic reaction), Cough, asthma, low fever, hemoptysis, chest-pain

Oral Ingestion of filariform larva cause:

- ✓ **“Wanaka syndrome”** showing Nausea, vomiting, pharyngeal irritation, cough, dyspnea: - (*A.duodenale*)



Intestinal phase

- ✓ Adult worms attach to the mucosa and continuously move to new places to feed on blood thus worsen bleeding.
- ✓ 0.03 ml for *N. americanus* & 0.26 ml *A. duodenale*, up to 200 ml per day in heavy infections
- ✓ high worm loads (20 - 100 worms) coupled with malnutrition can
- ✓ lead to anemia, iron deficiency, weakness and fatigue in adults, stunt growth and development in children



✓ Clinical features and pathogenesis

Site	Symptoms	Pathogenesis
Dermal	Local erythema, macules, papules (ground itch)	Cutaneous invasion and subcutaneous migration of larva
Pulmonary	Bronchitis, pneumonitis and, sometimes, eosinophilia	Migration of larvae through lung, bronchi, and trachea
Gastro-intestinal	Anorexia, epigastric pain and gastro-intestinal hemorrhage	Attachment of adult worms and injury to upper intestinal mucosa
Hematologic	Iron deficiency, anemia, hypoproteinemia, edema, cardiac failure	Intestinal blood loss



✓ **Laboratory Diagnosis**

❖ Finding eggs in feces

- ✓ Usually in freshly passed feces by saline wet mount and/or better with formalin-ethyl concentration technique
- ✓ *A.duodenale* & *N.americanus* eggs are indistinguishable morphologically
- ✓ If >12 hours old, a larva may be seen inside the egg and the larva hatch If >24 hours old

❖ PCR, immunological or serological tests (IgE), Eosinophilia



✓ Treatment, Prevention and control

- ✓ Albendazole 400mg, Pyrantel pamoate, Mebendazole 500mg, Thiabendazole
 - ✓ If anemic: high protein diet supplemented with ferrous sulphate, folic acid and vitamin B12
-
- ✓ Sanitary disposal of night soil
 - ✓ Individual protection (shoe wearing)
 - ✓ Health information and treat the patients and carriers.
 - ✓ Cultivate personal and environmental hygienic habits

✓ *Strongyloides stercoralis*

✓ **Common name:** Dwarf thread worm

Epidemiology

- ▶ Found worldwide with an estimated 50 -100 million cases
- ▶ Favors warmer tropical and subtropical climate
- ▶ In Ethiopia not highly prevalent in most areas and is found in the same geographical areas with hookworm
- ▶ The infection is rare or absent in many arid lowland areas like Ogaden and pastoral areas in the Awash Valley

Strongyloides stercoralis cont'd...

- ▶ Human is the definitive host, but can infect primates and dogs
- ▶ Worms has both free living and parasitic generations
- ▶ No parasitic males and parasitic females produce egg parthenogenically
- ▶ Can develop in to free living generation in the soil out side the human host
- ▶ Females reside in the small intestine and has internal auto infection

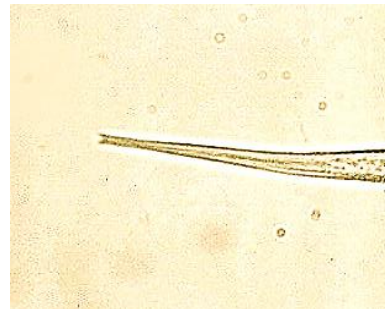
Habitat

- ▶ **Parasitic Adult females:** buried in the mucosal epithelium of the small intestine of man
- ▶ **Rhabditiform larvae:** Passed in the faeces and external environments
- ▶ **Filariform larvae:** soil and water, the infective stage
- ▶ **Free living male and female:** on external environment
- ▶ **Egg:** laid in the sub mucosa of small intestine

Strongyloides stercoralis cont'd...

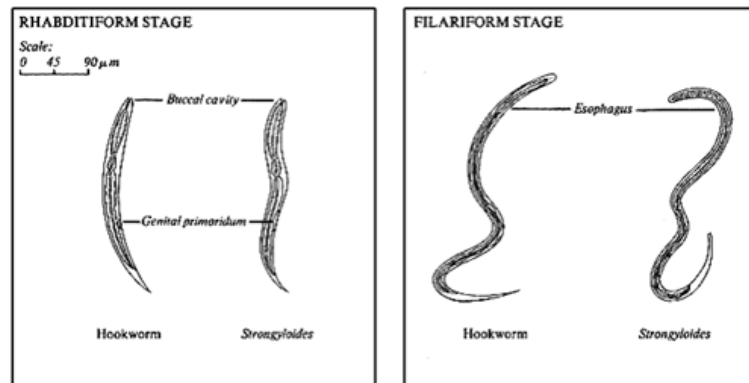
Morphology

- ▶ The size and shape of the worm are dependent on whether it's parasitic or free-living on external environment
- Free-living females – 1 mm X 60 μ m
- Parasitic females – 2.2 mm X 45 μ m
- Eggs – 55 μ m X 30 μ m: as soon as they are laid in sub mucosa, the rhabditiform larvae hatches



Strongyloides stercoralis cont'd...

- ▶ **Rhabditiform Larvae (non infective, diagnostic stage)**
- ▶ Size of 200-300 μ m long; 15 μ m thick & very actively motile
- ▶ Moderately tapered tail, short buccal cavity and esophagus
- ▶ **Filariform Larvae (infective stage, pathogenic stage)**
- ▶ About 600-700 μ m, Cylindrical esophagus and bifid tail end



✓ **Transmission**

- ▶ Skin penetration of filariform larva
- ▶ Ingestion of filariform larva thru contaminated food or water
- ▶ Transmammary & organ transplantation: rarely
- ✓ Autoinfection, the rhabditiform larva become infective filariform larva in the host tissue,
 - ✓ Penetrating intestinal mucosa (internal autoinfection) or perianal area (external autoinfection)

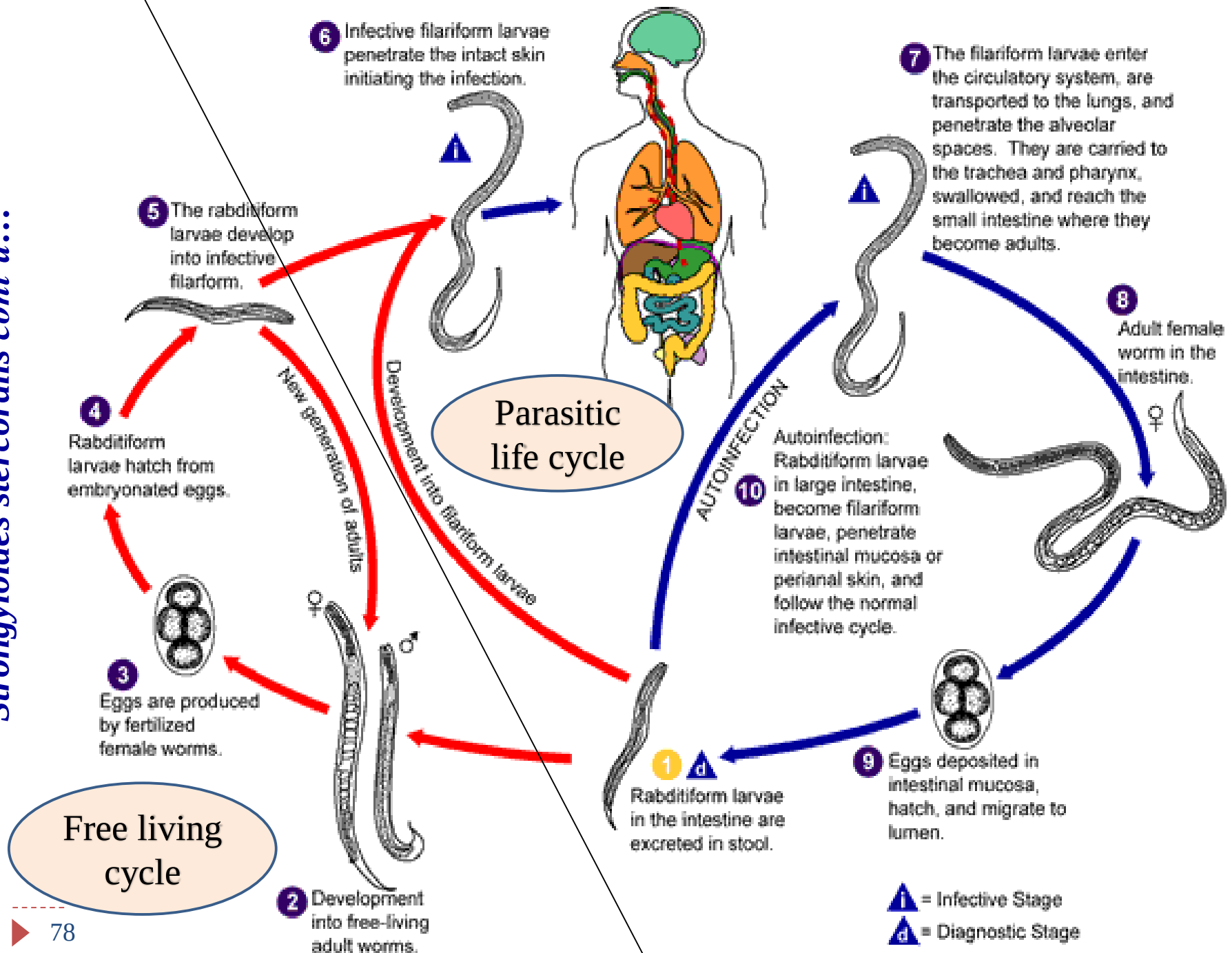
Life Cycle

- Complex, Two types of cycles exist:
- ✓ **Free-living (indirect) cycle**
- ▶ Rhabditiform larvae in stool → molts 4x → free living adult males and females produce eggs → rhabditiform molts 2x → Filariform larvae develop to free living adult males or females larvae ↓
(filariform larvae initiate parasitic life cycle)

Strongyloides stercoralis cont'd...

✓ Parasitic (direct) cycle

- ▶ Rhabditiform larvae in stool → molt 2x → develop to filariform → penetrate skin → lung → Alveolar space → bronchial tree → pharynx → swallowed → molt 2x → develop to adult female in small intestine
produce egg by parthenogenesis which yield rhabditiform larvae



✓ Clinical feature

- ▶ It is usually asymptomatic
- ▶ People with weaker immune systems are more susceptible
- ▶ **Cutaneous phase**
- ✓ Large number of larva produce itching and erythema at the site of infection within 24 hours of invasion
- ▶ **Pulmonary phase**
- ✓ The migratory larva in the lung producing bronchopneumonia and full blown pneumonitis

Strongyloides stercoralis cont'd...

▶ **Intestinal phase**

- ✓ Adult worms Invasion produce abdominal pain and mucus diarrhea, nausea vomiting & anemia

▶ **Hyper-infection syndromes due to hyper infection**

- ✓ Characterized by massive larval invasion of the lung or any other organ including CNS, which is fatal
- ✓ Occurs when the immune status of the host suppressed by either drugs, malnutrition or the concurrent diseases

✓ **Laboratory Diagnosis**

- ▶ Finding the larvae in faeces or in duodenal aspirates using direct or concentration method by microscopy
- ▶ Serological techniques
- ▶ Nematode Larva culture techniques (harada-mori filter paper, agar plate)
- ▶ Examination of serial samples may be necessary because direct stool examination is relatively insensitive

Treatment

- Ivermectin oral 200µg/kg daily for 2 days
- Albendazole 400 mg daily for seven days

Prevention and Control

- Proper disposal of feces in latrine, Avoid use of night soil as a fertilizer
- Wearing protective footwear, Treating infected individual, Health information

✓ *Trichuris trichiura*

- ▶ Whipworm: Due to the whip-like body form

Epidemiology

- ▶ Infections more frequent in tropical areas, poor sanitation practices and among children
- ▶ ~ 1 billion people are infected annually, prevalence of 11.6% in SSA countries
- ▶ In Ethiopia: Prevalence varies in localities, 1/4th are infected

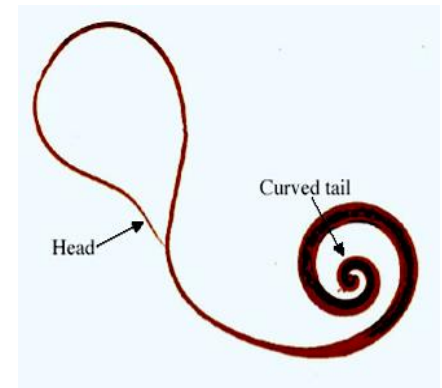
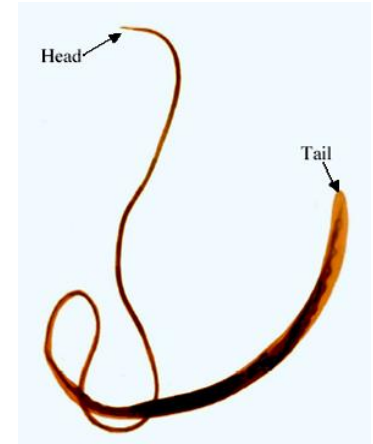
Trichuris trichiura cont'd...

Habitat

- ✓ **Adult:** large intestine (caecum) and appendix
- ✓ **Eggs :** In the feces, not infective when passed

Morphology

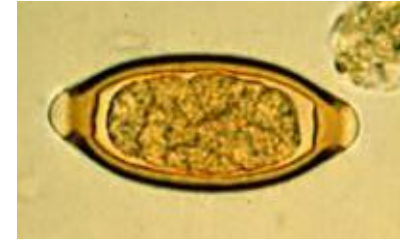
- ✓ **Adults:** whip-like shape, anterior $\frac{3}{5}$ th of the worm resembles a whip & the posterior $\frac{2}{5}$ th are thick
- ✓ **Male :** Size 30-45 mm , coiled tail
- ✓ **Female:** 35-50mm, straight thick tail



Trichuris trichiura cont'd...

Egg:

- ▶ **Size:** 50-54µm
- ▶ **Shape:** "tea tray eggs" or barrel - shaped with a colorless protruding mucoid hyaline plug at each end
- ▶ **Shell:** fairly thick and smooth, with two layers & bile stained
- ▶ **Color:** yellow brown
- ▶ **Content:** un segmented central granular mass ovum



Trichuris trichiura cont'd...

✓ **Transmission**

- ✓ Ingestion of embryonated egg in contaminated water, food or from contaminated hand

✓ **Life Cycle**

- ▶ Embryonated eggs are swallowed - larvae released into the upper duodenum (intestinal microflora as hatching stimulus)
- ▶ Then attach themselves to the villi of small intestine or invade the intestinal walls

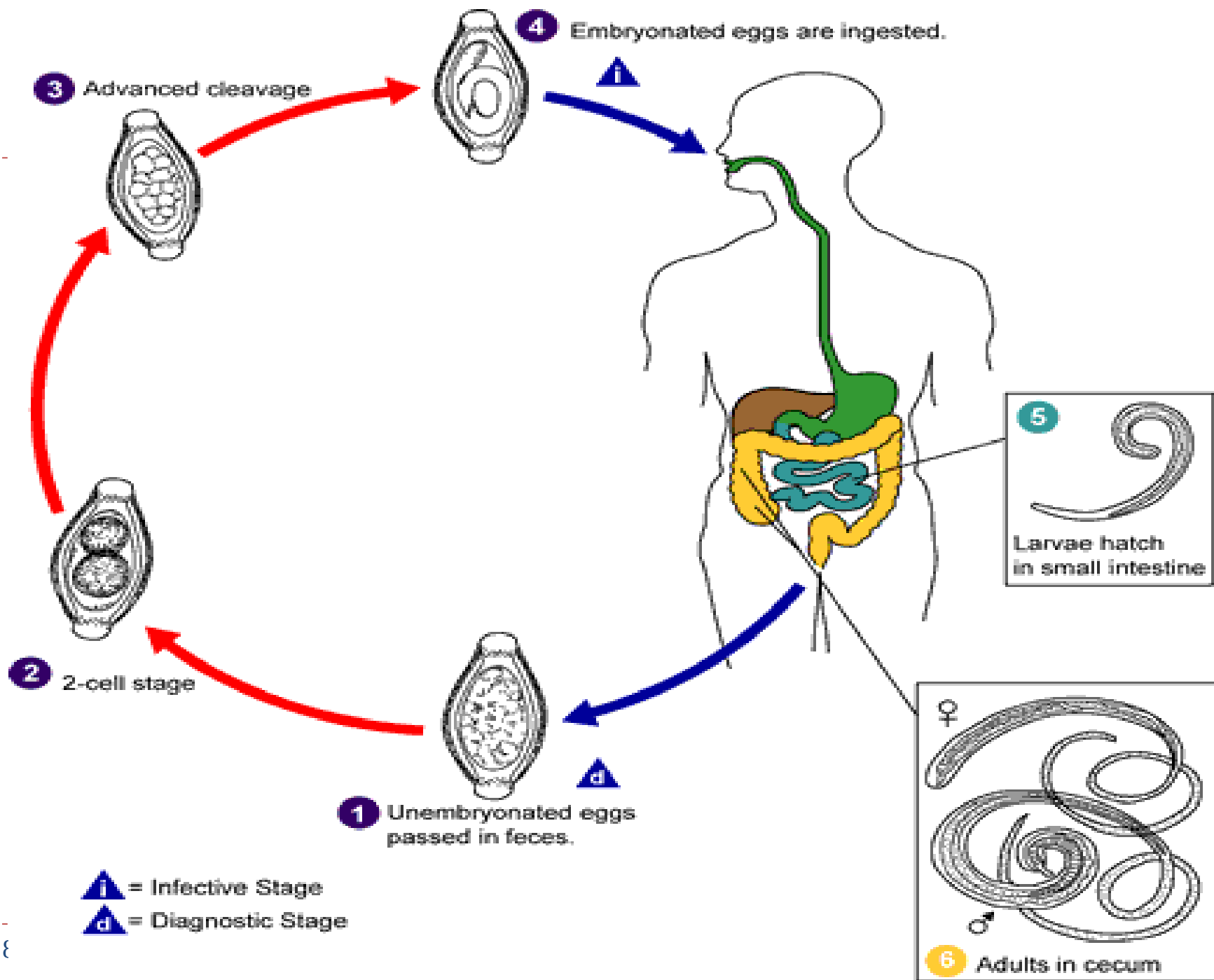
Trichuris trichiura cont'd...

- ▶ After 3-10 days they move down to the caecum & ascending colon where they mature into adult worms
- ▶ The adult worms are fixed with the anterior portions threaded into the mucosa
- ▶ The females begin oviposit 60 to 70 days after infection & shed 3,000 - 20,000 eggs per day
- ▶ The life span of the adults is about 1-5 years

Trichuris trichiura cont'd...

- ▶ The unembryonated eggs are passed with the stool
- ▶ Mature within 3wks of being deposited in soil.
- ▶ Require a warm, moist environment with plenty of oxygen to ensure embryonation
- ▶ The embryonated eggs are extremely resistant to environmental conditions

Trichuris trichiura cont'd...



▶ **Pathogenesis & Clinical features**

- ▶ Are largely determined by the worm burden:
- ▶ <10 worms are asymptomatic (99% asymptomatic)
- ▶ Heavy worm burden:
 - ▶ Mechanical damage to the intestinal mucosa
 - ▶ Chronic profuse mucus and bloody diarrhea with abdominal pains and edematous prolapsed rectum

Trichuris trichiura cont'd...

- ▶ Anaemia from blood loss & iron deficiency, malnutrition, weight loss,
- ▶ growth retardation, impaired mental development and cognitive function

Each adult worm sucks about 0.5µl of blood per day



Prolapsed rectum with
adult *Trichuris trichiura*

Trichuris trichiura cont'd...

Laboratory diagnosis

- ▶ Microscopic examination of barrel shaped egg in feces
- ▶ Sigmoidoscopy may enable visualisation of threaded worms

Treatment

- ✓ Mebendazole 100mg as first choice, alternatively albendazole 400mg, ivermectin 200mg for three days and treat the iron deficiency anaemia
- ▶ **Prevention and Control**
- ▶ Proper faecal disposal, Wash fruits and vegetables before consumption, Personal hygiene, Treatment of infected persons

✓ *Enterobius vermicularis*

- ✓ Common name: Pin Worm or Threadworm or Seat worm

Epidemiology

- ✓ Occurs world-wide in those groups living together
- ✓ Children <14 years are more commonly infected than adults especially during the “teething stage.”
- ✓ Eggs can spread throughout a house and difficult to eliminate
- ✓ In Ethiopia, a prevalence rate of 0.3-2.9% has been reported

Enterobius vermicularis cont'd...

Habitat

- **Adult:** Caecum & appendix
- **Gravid female:** Caecum & rectum
- **Eggs:** deposited on perianal skin & occasionally in feces

Morphology

- **Adults:** Color: small yellow white
- **Male:** Size 2-5mm Coiled tailed
- **Female:** 8-13mm, thin pointed tail



Female worm

Swollen cuticle at anterior end
Prominent esophageal end bulb

Enterobius vermicularis cont'd...

Egg

- ▶ **Size:** 50-60µm
- ▶ **Shape:** oval but flattened on one side, rounded on the other
D shaped
- ▶ **Shell :** Smooth and thin but with double shell
- ▶ **Content:** either a small granular mass or a small curved up larvae



Enterobius vermicularis cont'd...

Transmission

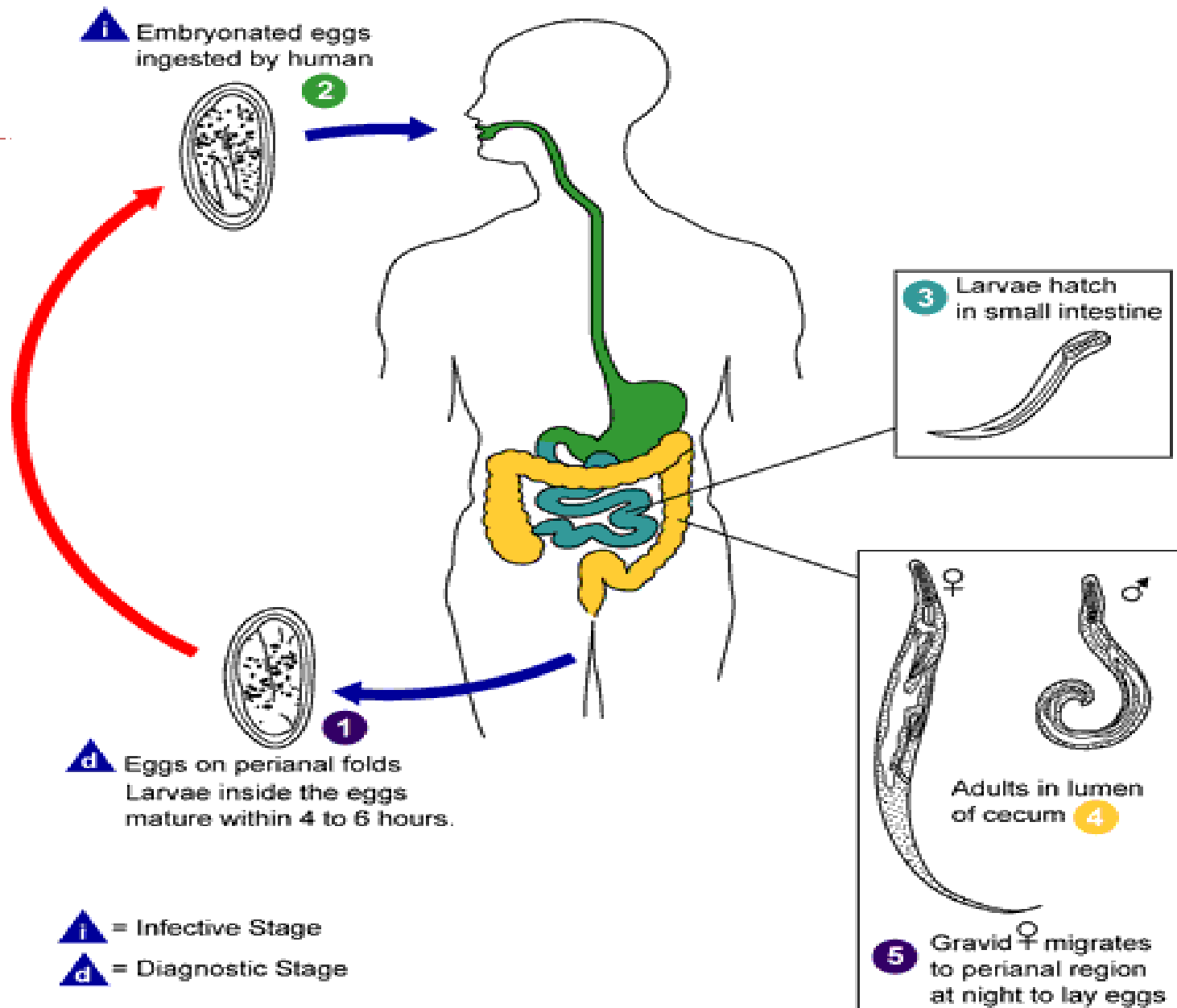
- ✓ **Ingestion** or air **inhalation** of eggs, viable for 20 days in env't
- ✓ **Auto-infection**: eggs transferred to mouth from fingers during anal itching
- ✓ **Retro-infection**: Migration of newly hatched larvae back into the rectum from the anal area

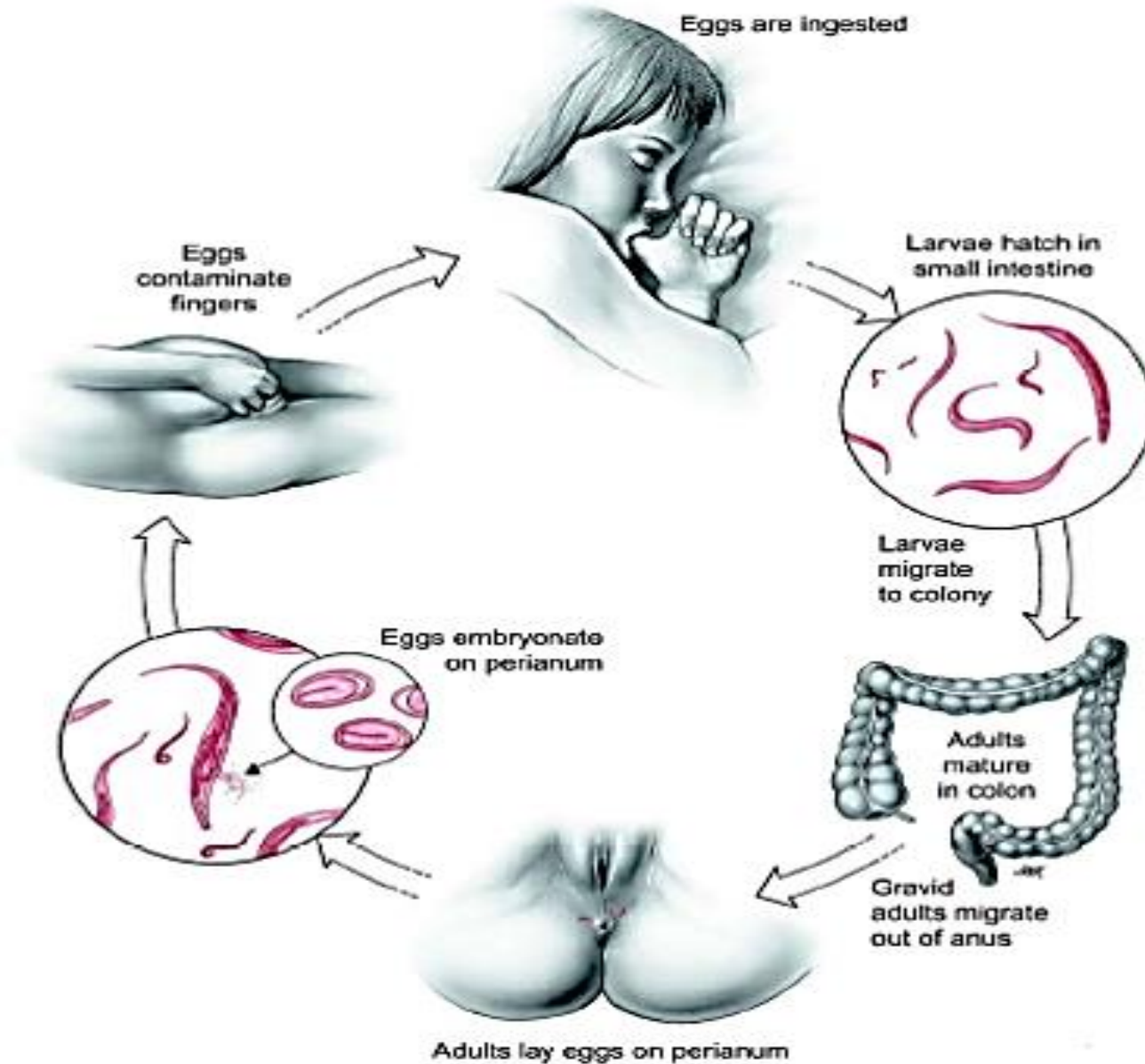
Life cycle

- ✓ Ingestion of embryonated eggs, carried on fingernails, clothing, bedding or house-dust and eggs hatch in stomach

Enterobius vermicularis cont'd...

- ▶ Larvae migrate to caecal region where they mature into adults & copulate – the adults lifespan is about 2 months
- ▶ Gravid females migrate **nocturnally** into the anal area
- ▶ Oviposition of 10,000 -15,000 eggs /day at the perianal area within 1 months of exposure
- ▶ Within 4-6 hours being laid the egg contain infective larvae
- ▶ Perianal itching from the eggs Induces scratching, and newly hatched larvae will migrate back into the rectum



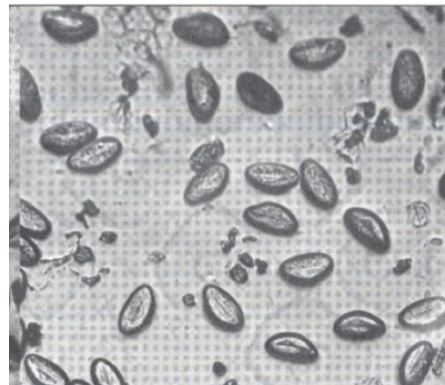


✓ **Clinical features**

- Nocturnal anal pruritus (an intense itching sensation) and thus sleeplessness
- The cause of irritation may be related to the intensity of the infestation, and/or an allergic reaction to parasite
- Vulvovaginitis, urethritis in girls when migrating worms
- Abdominal pain or appendicitis due to the worms are rare

✓ Laboratory Diagnosis

- ✓ Finding eggs by cellulose-tape slide test, “Scotch test” or anal swab or swab tubes method
- ✓ Pressing transparent adhesive tape on the perianal skin & examine the tape placed on a slide.
- ✓ Collect sample before defecation and washing, early in the morning



Enterobius vermicularis cont'd...

- ✓ Finding eggs in the feces as the eggs can be found less frequently (<10%)
- ✓ Occasionally eggs can be found in the urine or vaginal smears
- ✓ Finding of female worms from perianal skin during ano-rectal or vaginal examination



Treatment

- ▶ Mebendazole or Albendazole or Pyrantel Pamoate

Prevention and Control

- ▶ Treating all family members in which infection has occurred
- ▶ Wearing tight-fitting cotton pants to infected children
- ▶ Wash anal skin each morning, cloths worn at night, hands after toilet and before eating
- ▶ Avoidance of putting fingers in the mouth & trimming finger

✓ Blood and Tissue nematodes

✓ General features:

- ▶ Adults live in the tissues of human (lymphatic system, subcutaneous tissues or muscle)
- ▶ Are long thread like worms
- ▶ Requires **two host** to complete their life cycle
- ▶ Females are **viviparous**, larvae hatch in the uterus which is known as **microfilaria (L₁)**

➤ **Microfilariae**

- Are small, slender, motile forms
- requires blood sucking **insects (IH)** to develop to infective form (**L₃**)
- No reproduction in the vector, rather development

✓ **Three families/ groups**

1. Family **filariidae** (filarial worm)

- ▶ *Wuchereria bancrofti*, *Brugia malayi*, *Brugia timori*, *Loa loa*,
Onchocerca ovolvulus

2. Family **trichineloidae**

- ▶ *Trichinella spp*

3. Family **dracunculidae** (guinea worm)

- ▶ *Dracunculus medinensis*

▶ **Flarial worm causes 3 main diseases**

- ▶ Lymphatic filariasis (Elephantiasis), Loasis, Onchocerciasis (river blindness)

✓ **Lymphatic Filariasis**

- Disease caused by filarial worms living in the human lymphatic system
- **Causative agents**
 - *Wuchereria bancrofti*, *Brugia malayi*, *Burigia timori*

✓ Impact

- ✓ **Psycho-Social consequences:** It is one of the most debilitating and disfiguring diseases of the world
- ✓ **Disfigurement** of the limbs & genitals leads to Stigma, Psychological trauma, Sexual disability
- ▶ The disease **impedes** mobility, travel, education & employment opportunity, Marriage prospect

